



How is biodiversity changing in response to ecological restoration in terrestrial ecosystems? A meta-analysis in China

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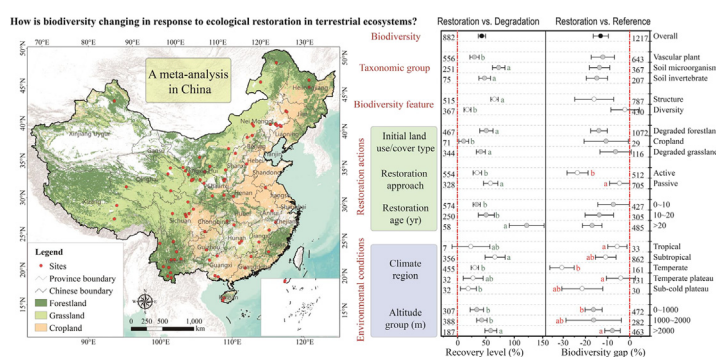
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HIGHLIGHTS

- Restoration enhanced biodiversity in degraded ecosystems but couldn't recover to natural level.
- Restoration improves habitat condition more than vegetation coverage.
- Impacts of restoration on biodiversity depended on restoration actions and climate conditions.

GRAPHICAL ABSTRACT



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ABSTRACT

Biodiversity is an important ecosystem characteristic, and is vital for maintaining ecosystem health and stability. However, biodiversity was often ignored in previous Chinese restoration planning and design due to its complex roles and the unclear mechanisms in providing human well-being. In order to evaluate the response of biodiversity to ecological restoration in terrestrial ecosystems, we assembled biodiversity in different metrics and different organisms and generated a large dataset comprised 2099 observations from 103 published studies to conduct a meta-analysis in China. Our results revealed that the biodiversity of restored ecosystem increased by 43% compared with degraded state, but it was difficult to recover to the natural level across the whole China. The gap between restored and natural ecosystems was about 13%. Ecological restorations have contributed not only to increasing vegetation coverage but also to improving soil environment and habitat quality. The recovery levels of vascular plant, soil microorganism and soil invertebrate were 30%, 73% and 48%, respectively. Biodiversity recovery would be better reflected in enhancing the structure feature (65%) such as plant height and density rather than the diversity feature (18%) such as diversity indices of Shannon and Simpson. Moreover, the response of biodiversity to ecological restoration varied with restoration actions (i.e., initial land use/cover type, restoration approach and restoration age), and the interaction effects among restoration actions significantly impacted biodiversity recovery. Passive approach performed better than active approach for biodiversity recovery. Meanwhile, the magnitude and direction of the impact of ecological restoration on biodiversity greatly altered with environmental conditions (i.e., climate condition and altitude). Our findings could facilitate priority setting and selection of treatment methods for biodiversity recovery during ecological restoration planning and assessment to ensure high effectiveness and sustainability.

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1. Introduction

Due to intensive human activities and dramatic climate changes, natural ecosystems have been severely degraded or damaged, leading to habitat fragmentation, ecosystem function degradation and ecosystem service loss (Benayas et al., 2009; Jackson and Hobbs, 2009; Allan et al., 2015). Ecological restoration, which aims to enhance ecosystem characteristics such as carbon sequestration, hydrologic regulation and species and landscape diversity enhancement (Stanturf et al., 2014; Kollmann et al., 2016; Deng and Shangguan, 2017), has been performed in degraded ecosystems at different spatial scales by implementing ecological restoration projects such as the Grain-to-Green Program (GGP) in China (Lamers et al., 2015; Deng et al., 2017a). Meanwhile, restoration actions are increasingly being supported by political decision makers and global policy commitments such as the Convention on Biological Diversity, which proposes a goal to protect at least 17% of terrestrial and 10% of marine areas in 2020 (Kullberg and Moilanen, 2014; Suding et al., 2015). Currently, improvements of ecosystem services are receiving more attention in restoration planning and design (Kollmann et al., 2016), however, biodiversity is often ignored because of its complex roles and the unclear mechanisms in providing human well-being, especially in previous Chinese restoration projects.

Biodiversity is perceived with varying connotations by different people (Maclaurin and Sterelny, 2008; Gaston, 2010). The definition proposed by the Convention on Biological Diversity, which is “the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part, including diversity within species, between species and of ecosystems”, is more commonly used to describe the term biodiversity in ecological restoration (Purvis and Hector, 2000; Mayer, 2006; Odenbaugh, 2009). According to this definition, the term biodiversity is a general concept, and is the sum total of all biotic variations from the level of genes to ecosystems (Purvis and Hector, 2000; Feest et al., 2010). Biodiversity could be measured in many different ways, and biodiversity metrics depended on the investigated organisms and purposes (Mayer, 2006; Mooers, 2007). Although biodiversity has a multitude of facets that can be quantified, it is still difficult to be fully explained by a single measuring feature such as diversity indices, which is held by scientists as well as non-scientists (Mooers, 2007; Gaston, 2010; Jax and Heink, 2015).

There is a widespread assumption that ecological restoration will increase biodiversity in the degraded ecosystems (Benayas et al., 2009). Some field studies have used many diversity metrics to evaluate the alterations in biodiversity caused by restoration actions. However, field studies typically focus on such small spatial scales that these results are most likely to be affected by the site conditions and monotonous organisms, especially plants (Cardinale et al., 2012; Wardle, 2016). Due to the environmental backgrounds varying with sites, the results and interpretations obtained from these field studies are not consistent, and debates on the responses of biodiversity to ecosystem changes have been contentious and lively (Balvanera et al., 2006). Fortunately, the evidence provided by meta-analysis suggests that ecological restoration could enhance biodiversity, which could allow future restoration efforts for biodiversity to achieve high effectiveness and sustainability (Benayas et al., 2009; Vellend et al., 2013). The present meta-analysis studies focused on the alterations in some specific species groups caused by restoration actions in the long-term, and only collected several biodiversity metrics to quantify the effects (Worm et al., 2006; Mooers, 2007). However, biodiversity is a comprehensive variable and should not be captured by several biodiversity metrics, and its changes should not be represented by the responses of several species groups to ecological restorations at large spatial scales (Maclaurin and Sterelny, 2008; Feest et al., 2010). Since plants, soil microorganisms and invertebrates are the most monitored organisms in terrestrial ecosystems, some meta-analyses integrated them to evaluate the biodiversity level at a large spatial scale (Benayas et al., 2009; Barral et al., 2015; Ren

et al., 2017). Therefore, it is necessary to integrate biodiversity in different metrics and different organisms for assessing restoration success at the regional scale in the meta-analysis, which would inform recovery decisions at large spatial scales.

Over the past two decades, the Chinese government has launched many ecological projects and land-use policies to improve environmental conditions and habitat qualities for terrestrial ecosystems (Liu et al., 2003; Li, 2004; Long, 2014; Deng et al., 2017b), especially vegetation restorations such as the GGP, Three North Shelterbelt Project (TNSP) and Natural Forest Protection Program (NFPP) (Li, 2004; Deng et al., 2017a). Although these ecological projects have been performed on a larger scale with a long duration, most of them are specifically targeted to recover a (or several) ecosystem characteristic(s), leading to relatively simple structure and composition for restored ecosystems (Tang et al., 2006; Cao et al., 2011). For instance, GGP is specifically targeted to prevent soil erosion by converting degraded croplands to forests, shrubs or grasslands (Deng et al., 2017a), but inflexible regulations of afforestation and subsequent poor management generate low diversity ecosystems such as monotonous tree plantation (Cao et al., 2011; Deng et al., 2016). Meanwhile, some studies documented that the effects of land use/cover type and restoration age are vital on ecological restoration (Liu et al., 2003; Deng et al., 2017a, 2017b). Some ecological restoration studies have revealed the changes in ecosystem characteristics following restoration actions and quantified the relationship between biodiversity and ecosystem services (Barral et al., 2015; Ren et al., 2017). China is a vast country with the complex physical environment, and these projects have directly provided raw materials and improved human living conditions by enhancing ecosystem services (Chazdon, 2008; Kollmann et al., 2016). However, the responses of biodiversity on ecological restorations and the biodiversity level of restored ecosystems in China remain uncertain and rarely quantified.

In this study, we used a standardized procedure to collect biodiversity observations from published ecological restoration studies in Chinese terrestrial ecosystems, and then quantified the recovery levels of biodiversity for overall and three taxonomic groups by a meta-analysis. To ensure suitable baselines for examination of restoration sequence, we separated observations into two comparisons of restoration vs. degradation and restoration vs. reference. The former would evaluate the actual recovery level of biodiversity, and the latter would assess the biodiversity gap between restored and natural ecosystems. Typical biodiversity metrics in published field studies included key biomass indicators (such as plant height and density) and diversity indices (such as Shannon and Simpson), and were reclassified into structure and diversity features. Also, we collected the potential influential factors of restoration actions and environmental conditions to analyze the effects of these factors on biodiversity recovery. We raise the following research questions: (1) What is the biodiversity level of restored ecosystem compared with degraded or natural state? (2) Which taxonomic group is most supported by restoration? (3) Which feature (structure or diversity) recovers more drastically? (4) How restoration actions and environmental conditions effected on the recovery level of biodiversity? This study can contribute to empowering ecological projects towards future biodiversity recovery by policy makers and researchers, and it could provide a sustainable solution of biodiversity recovery by combining restoration actions and environmental conditions.

2. Materials and methods

2.1. Literature search and data extraction

To identify quantitative studies evaluating the effects of ecological restoration on biodiversity in terrestrial ecosystems, we performed a systematic search of the peer-reviewed literature from the Web of Knowledge (www.isiwebknowledge.com) and the China National Knowledge Infrastructure (CNKI, <http://www.cnki.net/>). We searched these databases on 26 October 2017 with no restriction on publication

year, using the following search term combinations: (restor* OR recreat* OR rehabilitat* OR recover* OR plant*) AND (biodiversity OR diversity*) AND (China OR Chinese*). The titles, abstracts and keywords of these published studies were examined to assess their potential for meeting the selection criteria. Only primary studies that satisfied the following criteria were included in the analysis: (1) the quantitative evaluation of variables related to biodiversity comparing restoration with either degradation or a reference (natural/undisturbed ecosystem) within the same assessment; (2) the means, observation numbers, and standard deviations (or standard error) of the chosen variables were directly provided or could be indirectly estimated from the reported data; and (3) the environmental conditions and restoration approaches were reported.

The geographical locations of field studies were extracted and mapped by ArcGIS 10.2. We collected biodiversity metrics such as abundance, richness, and vegetation cover. Field studies that involved more than one restoration action and plant species (and/or ecosystem) were treated as multiple observations. We also extracted the restoration actions including initial land use/cover type, restoration approach and restoration age, and collected the environmental conditions including the climate region and altitude (Appendix S2).

2.2. Meta-analysis

2.2.1. Response ratios

To ensure suitable baselines for examination of restoration success, we separated observations into two comparisons of restoration vs. degradation and restoration vs. reference. The former would evaluate the actual recovery level of biodiversity, and the latter would be used to assess the biodiversity gap between restored/present and the natural/undisturbed ecosystems.

In order to evaluate the biodiversity level of restored ecosystem when compared with degraded and natural state, we generated the overall biodiversity by integrating all collected biodiversity metrics in different groups of organisms. On one hand, we have converted all collected biodiversity metrics to effects by using meta-analysis, and the normalized effects allowed us to analyze all data together, despite the overwhelming heterogeneity of biodiversity metrics extracted from published studies (Balvanera et al., 2006). On the other hand, as a general concept, biodiversity is the sum total of all biotic variation (Purvis and Hector, 2000; Feest et al., 2010). Meanwhile, most organisms in our database are vascular plants, soil microorganisms and soil invertebrates, which are the most investigated organisms in field studies for detecting biodiversity change and assessing restoration success in terrestrial ecosystems (Barral et al., 2015; Ren et al., 2017). So we could use the normalized effects of these three taxonomic groups to calculate the overall biodiversity, despite the heterogeneity of organisms within and between species.

In this study, the natural logarithm of the response ratio (lnRR) (Gurevitch and Hedges, 1999; Hedges et al., 1999) was chosen as the effect of ecological restoration on biodiversity and calculated by Eq. (1).

$$\ln RR = \ln \left(\frac{X_e}{X_c} \right) \quad (1)$$

where X_e and X_c are the means of a biodiversity metric in the experimental group (restored state) and control group (degraded or reference state), respectively.

The variance (v) associated with each lnRR was calculated by Eq. (2).

$$v = \frac{S_e^2}{n_e X_e^2} + \frac{S_c^2}{n_c X_c^2} \quad (2)$$

where X_e and X_c are the means of a biodiversity metric in the experimental and control groups, respectively, n_e and n_c are the corresponding

observation numbers, and S_e and S_c are the corresponding standard deviations.

Because the main assumptions are more likely to be satisfied with a weighted random-effects model when dealing with ecological data synthesis (Hedges et al., 1999), we fit the random-effects model with the DerSimonian-Laird estimator (Wallace et al., 2012) and calculated the weighted mean of the natural logarithm of the response ratio ($\ln RR_{++}$) by using the metafor package in R software to determine the overall effect of ecological restoration on biodiversity. Moreover, the weight associated with each lnRR was estimated as the reciprocal of the variance of effect so that studies with more replication were counted more heavily, and the $\ln RR_{++}$ was calculated using Eq. (3).

$$\ln RR_{++} = \frac{\sum_{i=1}^k w_i \ln RR_i}{\sum_{i=1}^k w_i} \quad (3)$$

where k is the number of observations and w is the weight of each lnRR and equals the reciprocal of the variance ($1/v$).

We converted the $\ln RR_{++}$ into the change percentage (A) to estimate the recovery level or biodiversity gap by Eq. (4), and the recovery effect or biodiversity gap was considered significant if the confidence interval (CI) of the change percentage at the 95% level did not overlap with zero (Koricheva et al., 2013).

$$A = (e^{\ln RR_{++}} - 1) \times 100\% \quad (4)$$

where $\ln RR_{++}$ is the weighted mean of the natural logarithm of the response ratio.

2.2.2. Publication bias analysis

To check for publication bias, we generated funnel plots (Koricheva et al., 2013). The plots generated roughly funnel-shaped distributions in both comparisons of restoration vs. degradation and restoration vs. reference (Fig. S1), which indicated that there was little publication bias in our meta-analysis from subjective and quantitative perspectives.

2.2.3. Sensitivity analysis

In addition, many published studies characterized biodiversity using more than one metric, and thus, multiple observations from one field study might not be independent. To test for this possible pseudo-replication problem, we randomly selected one effect or several effects for each field study and assessed the $\ln RR_{++}$ for the random dataset and the whole dataset (Gurevitch and Hedges, 1999; Koricheva et al., 2013). The CIs of $\ln RR_{++}$ at the 95% level for the random dataset overlapped with those for the whole dataset. The ANOVA did not pass F test in the comparisons of restoration vs. degradation (Table S1) and restoration vs. reference (Table S2), which revealed that the inclusion of multiple observations from one paper did not bias our meta-analysis, suggesting our results were reliable.

2.2.4. Subgroup analysis

To identify potential sources of the heterogeneity of biodiversity recovery, we performed the following series of subgroup analyses (Appendix S2): (1) taxonomic groups: vascular plant, soil microorganism and soil invertebrate; (2) biodiversity features: structure and diversity; (3) initial land use/cover types: degraded forestland, cropland and degraded grassland; (4) restoration approaches: active restoration and passive restoration; (5) restoration age groups (unit: year): 0–10, 10–20 and >20; (6) climate regions: tropical, sub-tropical, temperate, temperate plateau and sub-cold plateau; (7) altitude groups (unit: m): 0–1000; 1000–2000 and >2000.

2.3. Statistical analysis

We used one-way analysis of variance (ANOVA) to identify whether biodiversity change percentages differed among the levels of subgroup. When testing whether homogeneity of variance was passed and significance was observed at $p < 0.05$, a least significant difference test (LSD) was used to compare the biodiversity recovery levels for multiple treatments. Meanwhile, multi-way ANOVA was used to test whether the interactions among restoration actions (initial land use/cover types, restoration approaches and restoration age groups) were significant for biodiversity recovery. We employed Spearman's rank correlation analysis to examine the relationships between the lnRRs and the continuous variables (restoration age and altitude).

3. Results

3.1. Literature review

All of the available peer-reviewed publications were searched and resulted in a list of 3872 studies from the Web of Knowledge and 1440 studies from the CNKI. According to the criteria for the meta-analysis, we selected 103 studies finally (Appendix S1, Sheet 1). The database included 2099 observations of 882 were in the comparison of restoration vs. degradation and 1217 were in the comparison of restoration vs. reference (Appendix S1, Sheet 2). In the former comparison, there were 556, 251 and 75 observations for vascular plant, soil micro-organism and soil invertebrate respectively. In the latter, there were 643, 367 and 207 observations for vascular plant, soil microorganism and soil invertebrate, respectively.

Typical biodiversity metrics in field studies mainly were key biomass indicators (such as aboveground vegetation biomass, plant height and density) and some diversity indices (such as ACE, Shannon and Simpson), so these metrics were reclassified as structure and diversity features (Table 1). There were 424 and 458 observations for structure and diversity features in the comparison of restoration vs. degradation, and 647 and 570 observations for structure and diversity features in the comparison of restoration vs. reference.

Ecological restorations in field studies were commonly performed on the degraded vegetation of terrestrial ecosystems. In our database, there were 100, 1537 and 460 observations for cropland, degraded forestland and grassland, respectively. Ecological restorations reported in published studies were classified as active or passive restoration approaches (Table 2). Meanwhile, restoration ages ranged from several months to several decades, and the mean age was about ten years. 48% of restorations were less than ten years, and only 26% of restorations have been implemented more than twenty years.

The selected studies were conducted in 28 provinces, and 103 study sites were evenly distributed in China (Fig. 1). According to temperature and rainfall conditions, China is divided into five climate regions: tropical region (41 observations in 4 studies), sub-tropical region (1219 observations in 40 studies), temperate region (617 observations in 44 studies), temperate plateau region (164 observations in 10 studies)

Table 1
Reclassification of biodiversity metrics for structure and diversity features.

Biodiversity features	Observation number	Biodiversity metrics of published papers
Structure feature	1071	Aboveground biomass, Root biomass, DBH (Diameter at breast height), Plant height, Bacteria populations, Basal area, Canopy cover, Density, Litter cover/amount, Abundance, etc.
Diversity feature	1028	Alatalo index, Evenness index, Gleason index, H'gen index, Heip index, Margalef index, Maturity index, Pielou index, Shannon index, Simpson index, Richness index, etc.

Table 2
Summary of the most commonly used approaches for ecological restoration in vegetation of terrestrial ecosystems.

Approach	Observation number	Example
Active	1066	Planting tree (or grass) in the degraded cropland for soil and water protection; Restoring natural vegetation in ecologically sensitive areas; Vegetation restoration in mining areas; Artificial grassland establishment; Improving management measures (fertilizer, irrigation, forest thinning, transplanting, hallowing, reseeding, mixed-sowing, ploughing)
Passive	1033	Protecting existing natural forests from excessive cutting; Fencing/grazing exclusion for grassland; Abandoning cropland

and sub-cold plateau region (63 observations in 10 studies). The altitudes of study sites fluctuated between 0 and 5000 m. Meanwhile, 779 observations were conducted in 0–1000 m, 670 observations were conducted in 1000–2000 m, and 650 observations were conducted in >2000 m.

3.2. Impacts of ecological restoration on biodiversity

According to the biodiversity change percentages calculated by 882 observations, the overall biodiversity of restored ecosystem increased by 43% compared with degraded state (Fig. 2). However, it was 13% lower in restored ecosystem than in natural ecosystem according to 1217 observations of the comparison of restoration vs. reference (Fig. 2). Compared with degraded state, the biodiversity levels of restored ecosystem were higher for all taxonomic groups, and increased significantly by 30% for vascular plant, 73% for soil microorganism and 48% for soil invertebrate (Fig. 2). Moreover, the results of ANOVA documented that the recovery levels of soil microorganism and soil invertebrate were significantly higher than vascular plant. The biodiversity levels of restored ecosystem were substantially lower than natural ecosystem for three taxonomic groups, and the biodiversity gaps differed insignificantly among three taxonomic groups, and were fluctuated around 13% for them (Fig. 2).

Subgroup analysis and ANOVA revealed that the recovery levels significantly varied with two features in the comparison of restoration vs. degradation. Compared with degraded state, restored ecosystem presented increasing trends in two features, with a value of 65% for structure feature and 18% for diversity feature. In the comparison of restoration vs. reference, the gap of structure feature between restored and natural ecosystems was very small, while the gap of diversity feature in restored ecosystem was significantly far away from that in natural ecosystem (Fig. 2). However, according to the ANOVA, the gaps between restored and natural ecosystems differed insignificantly between two biodiversity features.

3.3. Effects of restoration actions

In the comparison of restoration vs. degradation, the biodiversity recovery level varied notably among different initial land use/cover types according to the results of ANOVA (Fig. 3a). The biodiversity level of restored state enhanced about 50% for restoration of degraded forestland and grassland, but only enhanced 11% for restoration of cropland. In the comparison of restoration vs. reference, the biodiversity gaps between restored and natural ecosystems were 14%, 11% and 6% for restoration of degraded forestland, cropland and degraded grassland, respectively (Fig. 3b).

In the comparison of restoration vs. degradation, the biodiversity recovery levels were 35% and 58% for active and passive restoration approaches, respectively. In the comparison of restoration vs. reference, the biodiversity gaps between restored and natural ecosystems were

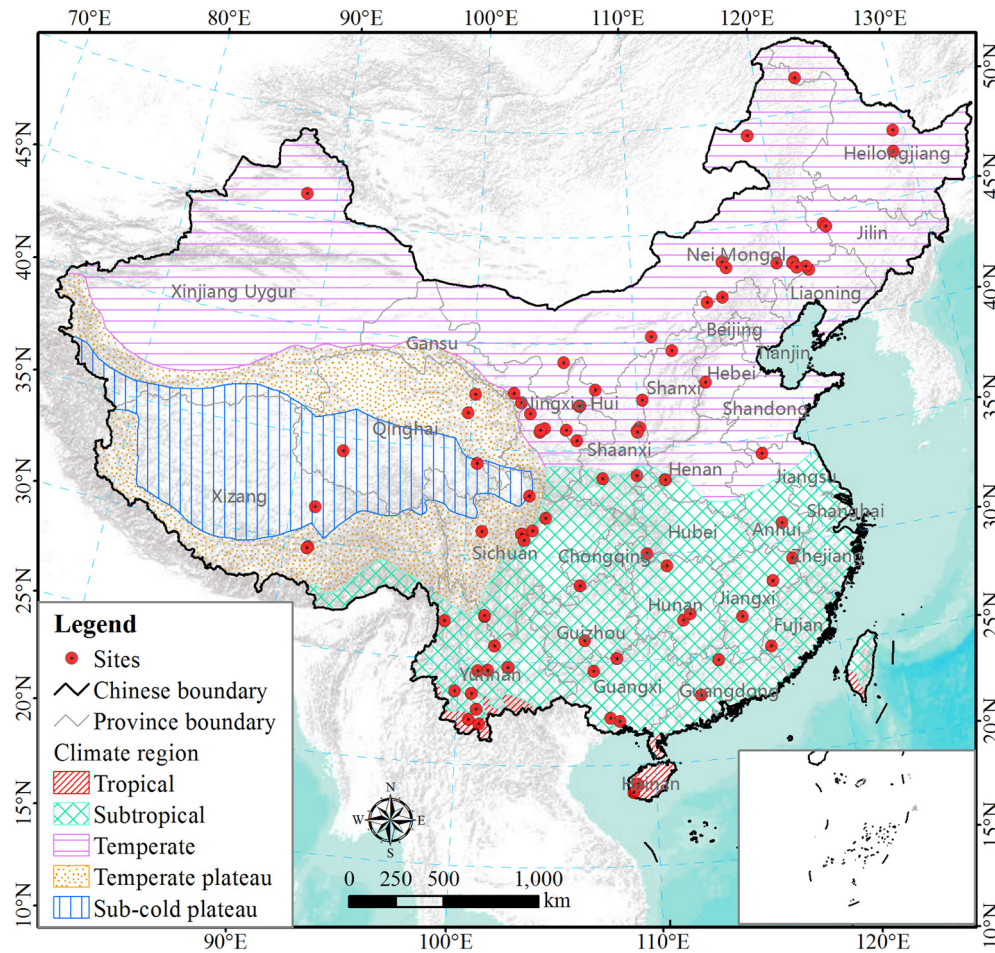


Fig. 1. The geographical distribution of field study sites included in our meta-analysis. Chinese climate region is provided by the Data Center for Resources and Environmental Sciences, Chinese Academy of Sciences (<http://www.resdc.cn>).

24% and 5% for active and passive restoration approaches, respectively. According to the results of ANOVA, the biodiversity recovery level of passive approach was significantly higher than that of active approach, and the biodiversity gap of passive approach was significantly smaller than that of active approach (Fig. 3).

Biodiversity markedly increased with restoration age in the comparison of restoration vs. degradation (Fig. 3a). Compared with degraded state, the biodiversity levels of restored state enhanced 34%, 51% and 122% for restoration age of 0–10, 10–20 and >20 years, respectively.

The biodiversity gaps between restored and natural ecosystems fluctuated around 10% for all three restoration age groups. Moreover, the biodiversity gaps between restored and natural ecosystems among these age groups were not significantly different because the ANOVA did not pass *F* test (Fig. 3b).

In addition, the multi-way ANOVA documented that the interactions among initial land use/cover type, restoration approach and restoration age significantly altered biodiversity recovery (Table 3). In the comparison of restoration vs. reference, the biodiversity gap between restored

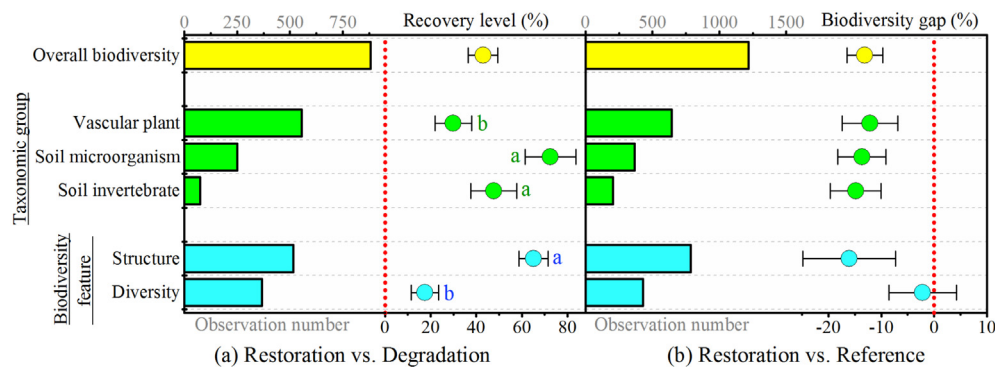


Fig. 2. Mean effects of biodiversity changes of overall, three taxonomic groups and two features in restored ecosystems with respect to (a) degraded ecosystems and (b) reference ecosystems. The observation numbers are illustrated by the bar charts. Bars extending from the means denote bias-corrected 95% confident intervals (CIs). If the 95% CI does not overlap with zero (red dotted line), the recovery effect is considered to be significant. Moreover, the different letters indicate a significant difference at $p < 0.05$ according to one-way analysis of variance.

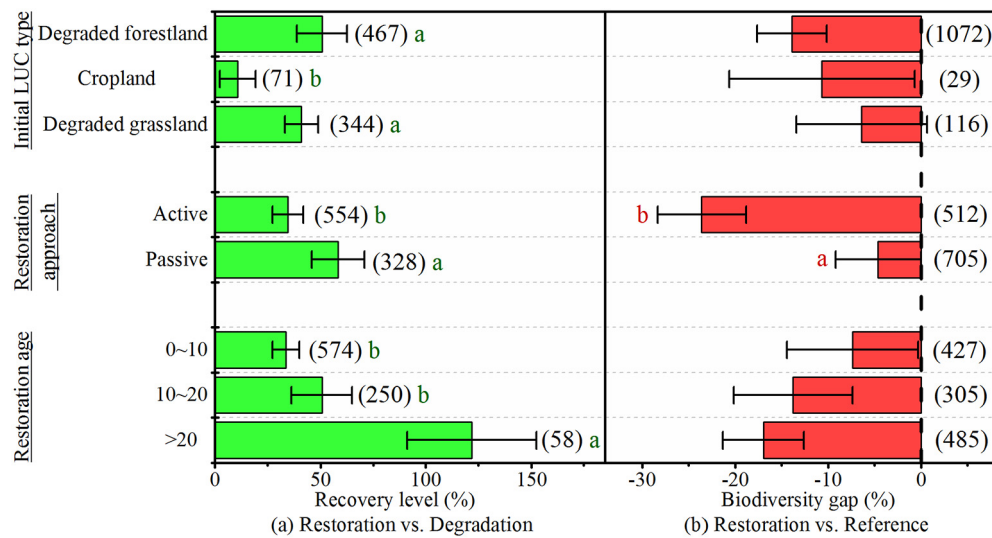


Fig. 3. Effects of restoration actions (initial land use/cover types, restoration approaches and restoration age groups) on biodiversity change percentages in the comparison of (a) restoration vs. degradation and (b) restoration vs. reference. Observation numbers are shown in parentheses. Bars extending from the means denote bias-corrected 95% confident intervals (CIs). The effect is considered to be significant when the 95% CI does not overlap with zero (black dotted line). Moreover, the different letters above the bars indicate a significant difference at $p < 0.05$.

and natural ecosystems was significantly influenced by the interaction between initial land use/cover type and restoration approach (Table 3).

3.4. Effects of environmental conditions

Compared with degraded state, the biodiversity level of restored state has significantly enhanced for most of the climate regions, except for the tropical region (Fig. 4a). The sub-tropical region had the largest increase in biodiversity (66%), followed by the temperate plateau region. In the comparison of restoration vs. reference, the biodiversity gap between restored and natural ecosystems was very small for the temperate plateau region. According to the results of ANOVA in both comparisons, the biodiversity recovery levels and the biodiversity gaps varied among climate regions (Fig. 4). The biodiversity recovery level of the temperate region was significantly lower and the biodiversity gap of the temperate region was significantly larger among five climate regions.

In the comparison of restoration vs. degradation, the biodiversity recovery levels were 34%, 43% and 59% for the altitude of 0–1000 m, 1000–2000 m and >2000 m, respectively. In the comparison of restoration vs. reference, the biodiversity gaps were 16%, 16% and 8% for these three altitude groups, respectively. According to the results of ANOVA, the biodiversity recovery level was significantly higher in >2000 m than in other altitude groups, and the biodiversity gap was significantly smaller in >2000 m than in 0–1000 m.

Table 3

The interaction effects of biodiversity recovery among restoration actions using multi-way ANOVA. Note: Biodiversity change in RD, the biodiversity recovery levels in the comparison of restoration vs. degradation; Biodiversity change in RR, the biodiversity gaps between restored and natural ecosystems in comparison of restoration vs. reference; ILUC, initial land use/cover type; Rapp, restoration approach; Rage, restoration age groups. ns, non-significant ($p > 0.05$); *, significant at $p < 0.05$; **, significant at $p < 0.01$.

Interaction effects	Sig. (p value)	
	Biodiversity change in RD	Biodiversity change in RR
ILUC × Rapp	0.000**	0.045*
ILUC × Rage	0.000**	0.484 ns
Rapp × Rage	0.000**	0.890 ns
ILUC × Rapp × Rage	0.000**	0.809 ns

4. Discussion

4.1. Response of biodiversity to ecological restoration

Our meta-analysis showed a strong positive effect of ecological restoration on biodiversity for degraded ecosystems, with biodiversity enhanced by an average of 43%, which was consistent with other meta-analyses conducted at the global scale (Benayas et al., 2009; Barral et al., 2015). We also compared the biodiversity gap between restored and natural ecosystems, and the biodiversity level was significantly 13% lower in restored ecosystem than that in natural ecosystem, which was in line with findings from other meta-analyses (Torralba et al., 2016; Ren et al., 2017). To the best of our knowledge, Chinese vegetation restorations mainly aim at a special environmental issue and has developed afforestation with just several tree species in restoration planning and design (Cao et al., 2011; Long, 2014; Deng et al., 2016), which led to the simple structure and lower diversity for restored ecosystems (Tang et al., 2006; Liu et al., 2014). Although ecological restoration contributed to controlling the loss of biodiversity, it was difficult to recover the natural level (the level of natural ecosystem).

Compared with degraded state, the recovery levels were significantly different among three taxonomic groups for restored ecosystem. Although these ecological restorations in Chinese terrestrial ecosystems were mainly focused on vegetation recovery (Liu et al., 2003; Long, 2014; Deng et al., 2017a, 2017b), the recovery levels of soil microorganism and soil invertebrate were significantly higher than that of vascular plant. Moreover, the recovery levels of soil microorganism and soil invertebrate were not significantly different. Some studies documented that soil microorganism and soil invertebrate were very sensitive to habitat changes and were effective indicators for monitoring environmental quality (Stephens et al., 2016; Lawes et al., 2017), implied that these ecological restorations have contributed not only to increasing the vegetation coverage (Zhu et al., 2018), but also to improving the habitat qualities of soil microorganism and soil invertebrate (Stephens et al., 2016). Moreover, vegetation restoration may contribute more to the soil environmental conditions (Stephens et al., 2016; Deng and Shangguan, 2017; Lawes et al., 2017). Our study could provide a scientific theoretical basis for ecological restoration to improve soil environmental conditions from the perspective of soil biodiversity. However, the collected observation number of soil invertebrate was fewer probably because of the difficulties in investigating and researching

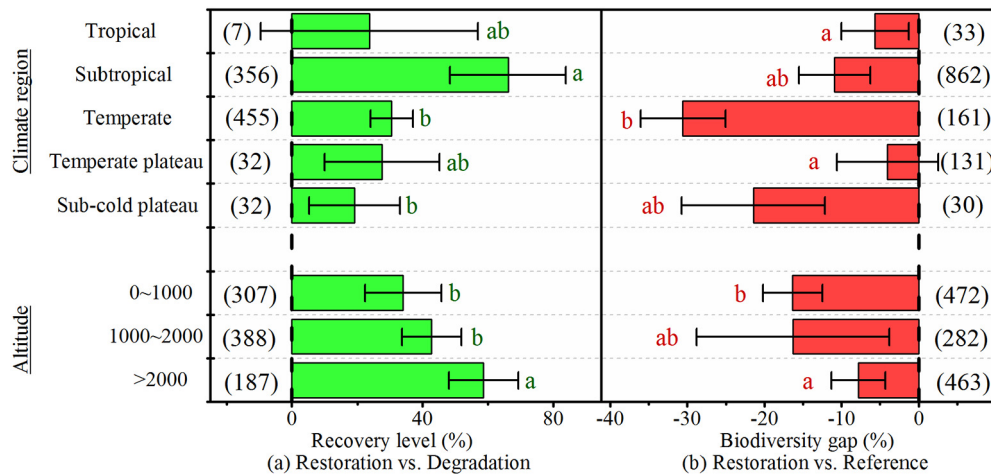


Fig. 4. Effects of environmental conditions (climate regions and altitude groups) on biodiversity change percentages in the comparison of (a) restoration vs. degradation and (b) restoration vs. reference. Observation numbers are shown in parentheses. Bars extending from the means denote bias-corrected 95% confident intervals (CIs). The effect is considered to be significant when the 95% CI does not overlap with zero (black dotted line). Moreover, the different letters above the bars indicate a significant difference at $p < 0.05$.

invertebrates and the strict data structure required for meta-analysis. This highlights an urgent need for biodiversity research on animals using more insightful experiments, which would allow meta-analyses to be performed with detailed, balanced and diverse datasets.

Based on the collected biodiversity metrics, we attempted to reclassify them into structure and diversity features (Table 1) and to analyze the effect of ecological restoration on them. In the comparison of restoration vs. degradation, the meta-analysis showed a larger increase in structure feature, which indicated that biodiversity recovery would be mainly reflected in enhancing the structure feature rather than the diversity feature in restored ecosystems. However, in the comparison of restoration vs. reference, the gaps between restored and natural ecosystems insignificantly differed with two biodiversity features. To the best of our knowledge, biodiversity features vary with the stage of recovery process, and the population mainly consists of the r-strategy species during the early recovery period but consist of the k-strategy species after many decades (Jackson and Hobbs, 2009; Kullberg and Moilanen, 2014). In our study, most of restoration ages were smaller in the former comparison (0–10 years accounting for 65%, >20 years accounting for about 7%), while most restoration ages were greater (>20 years accounting for about 40%) in the latter comparison. This may explain the inconsistency of two biodiversity features in the biodiversity recovery levels and biodiversity gaps.

4.2. Restoration actions and strategies to recover biodiversity

Theoretical and empirical studies have identified a variety of linkages between restoration actions and biodiversity changes (Benayas et al., 2009; Torralba et al., 2016). Our results supported suggestions that ecological restoration did not uniformly recover the mean levels of biodiversity across the three initial land use/cover types in the comparison of restoration vs. degradation. The degraded forestland showed the largest increment in biodiversity, followed by the degraded grassland. Moreover, the biodiversity recovery levels of these two initial land use/cover types had no significant differences. Afforestation of degraded forestland (or restoration of degraded grassland) may not significantly change ecosystem structures, but vegetation restoration of the cropland would significantly change biodiversity and ecosystem structures, such as plant species and plant cover, which may be attributed to the differences in biodiversity changes among initial land use/cover types and will largely contribute to ecological restoration performance in improving biodiversity at different sites (Bengtsson et al., 2005; Paillet et al., 2010).

According to the Chinese ecological projects, the restoration approaches of individual field studies were classified as active and passive restoration (Table 2). In general, active approach is more effective than passive approach for biodiversity recovery, as the former may accelerate the recovery process (Cao et al., 2011; Liu et al., 2014). However, our results documented that passive approach performed better than active approach in both comparisons, which was consistent with other meta-analyses conducted in forest ecosystems restoration (Ren et al., 2017). In addition, the interaction effects among initial land use/cover type, restoration approach and restoration age have significantly influenced biodiversity recovery (Table 3). Restoration approaches strongly depend on initial land use/cover types, degradation origins and restoration goals (Chazdon, 2008; Jackson and Hobbs, 2009; Suding et al., 2015). Active approach is always used in the regions with low vegetation coverage or non-vegetation coverage (bare lands), while passive approach is usually applied to the semi-natural vegetation (Liu et al., 2003). Moreover, some active restorations could introduce so many human activities that the species are very simple and homogenous at the beginning of recovery (Tang et al., 2006; Liu et al., 2014). There is considerable debate about whether and to what extent intensive interventions are necessary for facilitating the biodiversity recovery (Holl and Aide, 2011; Prach and del Moral, 2015). More and more studies have provided the evidence that passive restoration is an effective recovery strategy (Deng et al., 2016) and could achieve greater benefits for biodiversity with fewer costs than active restoration (Cao et al., 2011; Ren et al., 2017). However, it is necessary to compare the restoration approaches in different habitats and geographical regions to identify and selecting the effective methods of biodiversity recovery (Prach and del Moral, 2015).

As mentioned above, restoration age is crucial and essential in ecosystems restoration, driving both the vegetation community evolution and the habitat quality improvement (Jackson and Hobbs, 2009). According to the subgroup analysis in the comparison of restoration vs. degradation, the biodiversity recovery level presented a positive correlation with the restoration age (Fig. 4c), which was consistent with some field studies conducted in specific sites (Liu et al., 2003; Deng et al., 2017a, 2017b). We also employed Spearman's rank correlation analysis to further validate the relationships between biodiversity and restoration age (Fig. S2). However, the statistical results were not significant and the correlation coefficients were very small, which was consistent with other meta-analyses documenting that biodiversity recovery did not significantly correlate with restoration age (Barral et al., 2015; Ren et al., 2017). This probably because the effects were calculated by both the lnRRs and its variances in subgroup analysis, while the

correlation coefficients were calculated by only the lnRRs in correlation analysis. Therefore, the complex and usually nonlinear responses of ecological restoration to time are difficult to explain by the simple correlation analysis, and the subgroup analysis might be more understandable and reasonable for identifying the heterogeneity in the effect of restoration age on biodiversity recovery (Balvanera et al., 2006). Further research should examine this issue in more detail and at different scales.

4.3. Impacts of environmental conditions on biodiversity changes

The magnitude and direction of the impact of ecological restoration on biodiversity differed greatly based on climate conditions (Harris et al., 2006; Mantyka-Pringle et al., 2012; Allan et al., 2015). Compared with degraded state, the biodiversity recovery level of restored ecosystem was not significant in the tropical region, probably because the observation number of this region was fewer and the 95% CI of biodiversity change percentage was longer. Moreover, this region is too small to collect more available observations. There were many observations in the sub-tropical and temperate regions, which implied that the quantitative evidence of these two regions were more reliable. Meanwhile, the biodiversity recovery level was significantly lower in the temperate region than in other regions, and the biodiversity gap between restored and natural ecosystems was significantly larger in the temperate region than in other regions. The biodiversity recovery levels and biodiversity gaps significantly differed among the climate regions, probably because the warm and humid environmental conditions are so favorable for spontaneous succession that vegetation cover forms quickly, which provides a good habitat for biodiversity recovery (Kullberg and Moilanen, 2014; Suding et al., 2015).

Our meta-analysis documented that the biodiversity change percentages significantly varied among altitude groups in both comparisons. The biodiversity recovery level increased with altitude in the comparison of restoration vs. degradation, and the biodiversity gap between restored and natural ecosystems decreased with altitude. Meanwhile, the Spearman's rank correlation analysis also detected a significant positive relationship between the response ratio of biodiversity recovery and altitude (Fig. S3). Altitude is associated with many environmental factors such as temperature, rainfall, and radiation (Tang et al., 2006), and the altitudes varied widely from several meters to kilometers across large geographical ranges in our database. Therefore, the weak relationship (Fig. S3) may be caused by the complex and usually nonlinear responses of ecosystem restoration processes to altitude (Allan et al., 2015).

4.4. Limitations and uncertainties

Our study provided quantitative evidence that ecological restoration exerted significant impacts on biodiversity in terrestrial ecosystems. However, due to the inherent limitations of meta-analysis methodology and the lack of complete datasets at different scales, uncertainties still exist. First, as a common issue, the random-effects model with the observation-weighted approach may overestimate the effects of restoration due to the low variance in some observations (Gurevitch and Hedges, 1999; Koricheva et al., 2013). Nevertheless, the publication bias analysis (Fig. S1) and sensitivity analysis (Tables S1 and S2) suggested that this may not be the case for the large dataset used in our study. Second, since biodiversity is a complex topic and covers many aspects of biological variation (Mooers, 2007; Gaston and Spicer, 2013), there were some intersections of the reclassifications for those biodiversity metrics in our study. However, this study was the first attempt to reclassify the biodiversity metrics into structure and diversity features and analyze the responses of biodiversity features in China. We suggest that future studies would systematically explore the biodiversity features in different ways and reveal these biodiversity features changes following ecological restorations. Finally, although the literature search and the data selection were fruitful, we only focused on the restored

ecosystems implemented by the ecological projects in China (Fig. 1), which may be less able to explain these effects at global scale. Observations in field studies were fewer for soil microorganisms and soil invertebrates than for vascular plants. Moreover, although the key restoration practices and environmental conditions in each site were represented, there is a geographic bias in our pool of primary studies. In the regions with cold and arid climate conditions, field studies mainly concerned biodiversity recovery in grasslands, while in regions with more temperate and humid climate conditions, interest in forest recovery may be higher (Cao et al., 2011). We suggested that authors systematically provide detailed information on environmental conditions, habitat characteristics, species abundances and restoration history because it is important to control for the influence of these possible confounding factors in sampling strategies.

5. Conclusions

The meta-analysis showed a strong positive effect of ecological restoration on biodiversity for degraded ecosystem, but it was difficult to recover biodiversity to the level observed in natural ecosystem. Following vegetation restoration in Chinese terrestrial ecosystems, soil micro-organism presented the largest increase in biodiversity, followed by soil invertebrate, which indicated that ecological restorations have not only contributed to increasing vegetation coverage but also to improving soil habitat qualities. Moreover, typical biodiversity metrics included key biomass indicators (such as plant height and density) and diversity indices (such as Shannon and Simpson), and were reclassified into structure and diversity features. Biodiversity recovery would be better reflected in enhancing the structure feature than the diversity feature for restored ecosystems. The impacts of ecological restoration on biodiversity varied among different initial land use/cover types, and passive approach performed better than active approach for biodiversity recovery. Due to the lack of long-term observation data in present studies, further research should examine the impacts of restoration age on biodiversity recovery in more detail. In addition, the magnitude and direction of the impact of ecological restoration on biodiversity greatly differed with climate conditions. Our findings could facilitate priority setting and selection of treatment methods for biodiversity recovery during restoration planning and assessment for high efficiency, effectiveness, and sustainability.

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